

ANONYMOUS AFFILIATIONS

- HANDS-ON TUTORIAL -

How to Interact with the Agent-powered learning website

ANONYMOUS

Anonymous City, July 2026

Abstract

The AI-Agent system is an advanced, interactive educational platform specifically engineered to support students enrolled in the course Intelligent Systems. By integrating state-of-the-art generative AI capabilities with structured learning analytics, the platform serves as a personalized pedagogical companion that facilitates independent inquiry, conceptual mastery, and practical technical deployment. The system is designed to guide learners through customized academic trajectories based on their selected domains of specialization. Concurrently, the platform operates as an empirical framework for educational research; it systematically aggregates data regarding student progression, shifting attitudes toward AI technologies, and cognitive development. All diagnostic metrics are handled with strict anonymity, serving exclusively to optimize instructional design, improve curriculum efficacy, and advance the integration of artificial intelligence within higher technical education.

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Chapter 1

Overview

The AI-Agent system is an advanced, interactive educational platform specifically engineered to support students enrolled in the course Intelligent Systems. By integrating state-of-the-art generative AI capabilities with structured learning analytics, the platform serves as a personalized pedagogical companion that facilitates independent inquiry, conceptual mastery, and practical technical deployment.

The system is designed to guide learners through customized academic trajectories based on their selected domains of specialization. Concurrently, the platform operates as an empirical framework for educational research; it systematically aggregates data regarding student progression, shifting attitudes toward AI technologies, and cognitive development.

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Current website is hosting at Vercel for Interface, go to <https://support-learning.vercel.app>, currently available in a limited time.

Chapter 2

Student Account Registration and Profile Configuration

As you access the main web page, you need to create account to login and use the service.

The initial phase of onboarding requires students to generate institutional accounts using distinct authentication credentials (username and password).

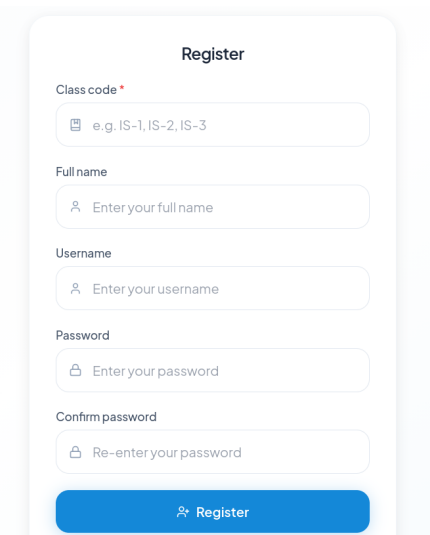
A mobile app registration form titled "Register". It contains five input fields: "Class code" with a red asterisk and a hint "e.g. IS-1, IS-2, IS-3"; "Full name"; "Username"; "Password"; and "Confirm password" with a hint "Re-enter your password". Each field has a small icon on the left. At the bottom is a blue "Register" button with a white user icon.

Figure 2.1: Registration form for creating a learner account.

Each student is required to fill in these fields:

- Class code: Each student must fill in their class code in the responding values: IS-1, IS-2, IS-3.

- Full name: Enter your fullname in English (e.g. Nguyen Van A).
- Username
- Password
- Confirm Password (same as your Password above, just to make sure the value is not misconfigured)

Upon creating account, user can now login with the responding username and password at the main login page.

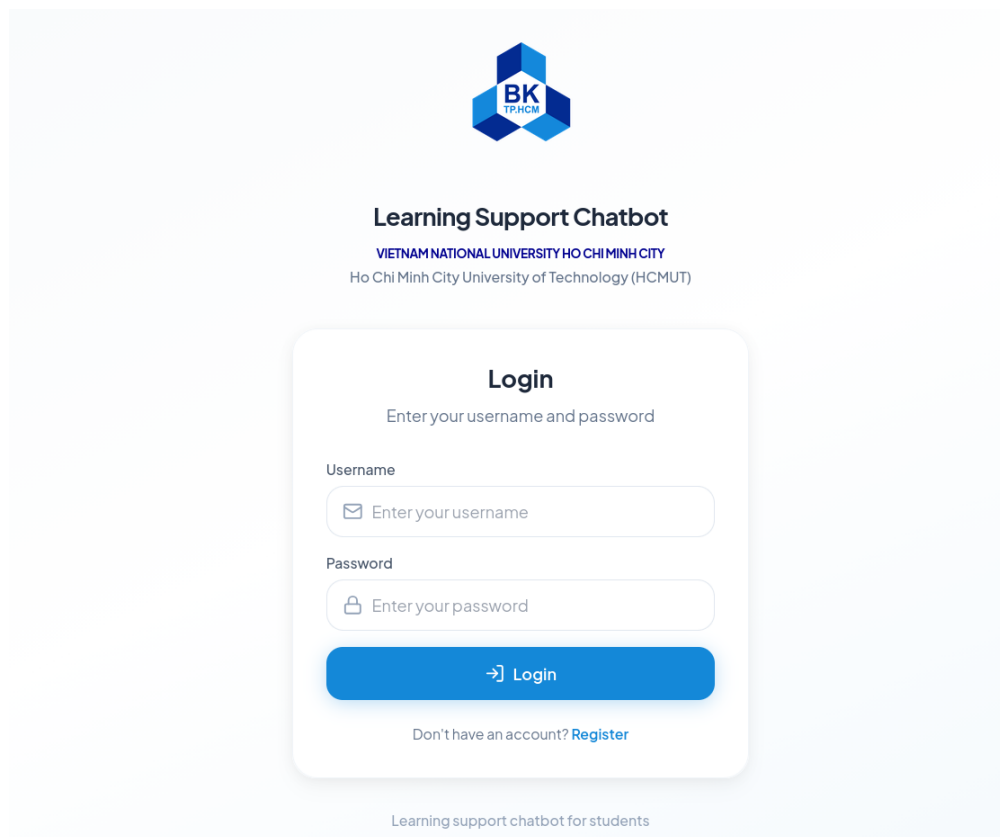


Figure 2.2: Login page for accessing the Learning Support Chatbot.

Later, individuals must proceed to the profile management interface to populate their personal and academic profiles. It is imperative that essential parameters are accurately inputted to ensure seamless system integration and data alignment, as agents use this information to help in understanding the context, the learners' study progress, and personalization.

These parameters must be filled on

- Email (School mail is preferred).
- Date of birth.
- Gender.
- Training program type.
- Student ID (must have).
- Faculty.
- Major.
- Subject of Interest.

The screenshot displays the 'Settings' page of a Learning Support Chatbot interface. The page is divided into three main sections: a left sidebar, a central form area, and a bottom section for language settings.

Left Sidebar:

- JOURNAL UPLOAD**
 - Journal upload
- CHAT CHANNELS**
 - Class IS-1
 - Chat with Agent (IS-1)
- Settings** (selected)
- Logout**
- Nguyen Van A** (Learner)

Central Form Area:

Account (Nguyen Van A) [Cancel]

Email: beta-tester1@secret.mail

Role: ☒ Learner

Student information

Email: beta-tester1@secret.mail Date of birth: 01/01/2000

Gender: Male Training program type: Undergraduated

Student ID: 2452141 Faculty: Computer Science and Engineering

Major: Computer Science Subject of interest: Machine Learning, Deep Learning

[Save]

Language

Choose interface language

Figure 2.3: Profile configuration page for student academic information.

Chapter 3

Tests

This evaluation phase is designed to rigorously assess specialized expertise across the two selected topics. The assessment instruments comprise a diverse matrix of examination items structured in strict alignment with the cognitive dimensions of Bloom's Taxonomy, spanning from lower-order to higher-order thinking skills: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating. Specifically, the evaluation metrics encompass core model objectives, fundamental mathematical formulations, and the statistical significance of key indicators. Furthermore, candidates are required to execute practical problem-solving tasks, perform feature analysis, critique system outputs, and propose innovative structural transformations.

In this course, we will have 3 kinds of tests:

- Pretest: To evaluate the students' initial knowledge, before the course and before using the Agent system. Which is also the required test to complete when you first login into your account.
- Posttest 1: To evaluate the students' learning progress right after completing the final task of the course, which is the final submission, including labs and essays.
- Posttest 2: To evaluate the students' knowledge after completing the course, to make sure using the Agent system helps learners on their overall performance and memory of the course.

Chapter 4

Weekly Progress Monitoring and Journal Submission

After completing the profile configuration step, students can access the progress-tracking module from the left-hand dashboard. This module helps students submit weekly learning evidence, monitor their own progress, and communicate with the Agent-powered support system.

The dashboard provides two primary functional modules:

- **Journal Upload:** The formal submission area where students upload weekly journals, final submissions, and post-test materials.
- **Chat with Agent:** The communication channel where students ask questions, request learning support, and receive real-time guidance.

The Journal Upload workflow includes the following required submissions:

- **Submission 1:** The first weekly journal for documenting the initial learning plan and early progress.
- **Submission 2:** The second weekly journal for reporting updated milestones and learning challenges.
- **Submission 3:** The third weekly journal for reflecting on technical development and concept mastery.
- **Submission 4:** The fourth weekly journal for summarizing near-final progress before the final deliverable.

- **Final Submission:** The complete final course submission, including required labs, reports, or essays.
- **Post-test 1 and Post-test 2:** The exit assessments used to evaluate learning outcomes after students use the Agent system.

Each weekly journal should include:

- the student's short-term learning plan for the current week;
- milestones completed during the week;
- goals and tasks planned for the next week;
- difficulties, blockers, or learning constraints;
- self-evaluation of strengths and areas that need improvement;
- questions that may require support from the AI Agent or instructor.

Students should follow this weekly reporting routine consistently because it supports:

- **Self-regulated learning:** Students can track what they have completed and what they still need to improve.
- **Progress monitoring:** The course team can observe learning development across the semester.
- **Personalized support:** The Agent system can provide more relevant guidance when it understands the student's learning context.
- **Research data collection:** Anonymous progress information can support future improvement of AI-assisted education.

Chapter 5

Requirements

To ensure systemic data alignment and maintain academic progression, students must follow the platform workflow in the correct order. These requirements help the system connect account information, profile metadata, test results, journal submissions, and learning support records.

The required workflow is as follows:

1. **Register an account:** Create an institutional learner account using a full name, username, password, and valid class code.
2. **Use the correct class code:** Enter one of the accepted class identifiers: IS-1, IS-2, or IS-3.
3. **Complete the Pre-test:** Submit the baseline survey before using the centralized chat channel or other core course utilities.
4. **Configure the profile:** Open the Settings page and save all required personal and academic information.
5. **Submit weekly journals:** Upload four sequential weekly journals to document milestones, plans, and learning constraints.
6. **Complete the Final Submission:** Submit the final course deliverables according to the required deadline.
7. **Complete Post-test 1 and Post-test 2:** Finish both exit assessments so the course team can evaluate learning outcomes.
8. **Report technical issues:** Document any technical anomaly through the reporting interface or official support channel provided on the website.

The Pre-test is mandatory because it establishes the learner's baseline profile. It is used to collect:

- demographic and academic background information;
- prior exposure to AI technologies;
- initial cognitive proficiency across the selected technical domains;
- individual readiness for AI-assisted learning;
- starting attitudes toward intelligent learning systems.

After completing the Pre-test, students must configure their profile with accurate information, including:

- Student ID;
- Faculty;
- Major;
- Course or Subject of Interest;
- Date of Birth;
- other required fields shown on the Settings page.

During the semester, continuous compliance requires students to:

- submit all four weekly journals on time;
- include milestones, self-regulation notes, and learning constraints in each journal;
- complete the Final Submission with the required labs, reports, or essays;
- complete both post-tests after the relevant learning stages;
- contact support immediately when technical issues prevent normal platform usage.

Following this sequence ensures that the AI Agent has enough context to provide personalized guidance and that the course team can evaluate student progress consistently.

Chapter 6

Conclusion

In conclusion, the AI-Agent system represents a systematic framework designed to bridge the gap between independent theoretical research and practical system implementation in the domain of intelligent technologies. By pairing intuitive, real-time conversational scaffolding with rigorous continuous assessment mechanisms, the platform cultivates a highly effective environment for self-regulated learning. The structural interplay between initial diagnostic pre-tests, recurring reflective weekly journals, and advanced exit post-tests ensures a transparent, empirical trajectory of student growth mapped directly against higher-order cognitive skills. Ultimately, this implementation not only enriches the individual learning experience by rendering complex technical architectures accessible, but it also generates critical, anonymized empirical insights. These insights will actively contribute to the broader academic discourse surrounding AI-assisted engineering education and inform future iterative improvements to high-tech university curricula.